

CHARLES AKINNURUN

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Education

Lagos State University of Education

Bachelor of Science Education in Computer Science

Grade: Second Class Upper (2:1)

lagos, Nigeria

Work Experience

Amazon Web Services (AWS)

Mar 2026 – Present

AI/ML Engineering Scholar

Remote

- Developed an AI productivity application that applied generative AI to automate common tasks, improving workflow efficiency and demonstrating practical application of AWS generative AI technologies.
- Implemented pre-trained CNN models (AlexNet, VGG, and ResNet) for dog and breed identification, comparing their classification accuracy and runtime to determine the best-performing architecture.
- Analyzed data using AI-powered workflows in Amazon PartyRock, translating user inputs and datasets into actionable insights through interactive prompts and application logic.

FlyRank AI

Jun 2026 – Present

Machine Learning Engineering Intern

Remote

- Developed and evaluated 3-5 machine learning models using python and scikit-learn, achieving up to 92% classification accuracy on the final validation dataset.
- Engineered an end-to-end machine learning pipeline that automated processing for 20,000+ records, reducing the manual data preparation by 70%
- Conducted 40+ machine learning experiments, optimizing hyperparameters to increase the F1-Score from 0.81 to 0.90.

Dev Weekends

Apr 2026 – Present

AI Engineering Fellow

Remote

- Engineered and deployed 5+ AI-powered applications using python and modern LLM frameworks, improving task automation and user interaction efficiency by p to 80%
- Built retrieval augmented generation (RAG) pipelines by integrating vector databases and embedding models, improving response relevance by 35% over baseline prompting
- Developed and evaluated 10+ prompt engineering strategies for large language models, increasing task completion accuracy from 82% to 94%.

CodeAlpha

Oct 2024 – Feb 2025

Data Science & Machine Learning Intern

Remote

- Developed a handwritten character recognition system using CNNs, achieving 94% accuracy in identifying diverse alphanumeric characters.
- Engineered a credit scoring model that accurately predicted loan default risk with an 88% AUC-ROC score, optimizing risk assessment protocols.

Featured Projects

AI Finance Agent | *Amazon Bedrock, Bedrock Agents, AWS Lambda, Amazon S3, Python, Pandas, AWS IAM*



- Developed an autonomous finance agent on Amazon Bedrock that writes and executes its own Python code to analyze uploaded spending data, iterating on user instructions and adding cross-session memory to deliver personalized, ongoing financial insights.

AI Email Router | *Bedrock Flows, Bedrock Prompt Management, Bedrock Guardrails, Foundation Models, AWS Console*



- Engineered a visual, node-based AI workflow using Amazon Bedrock Flows to automatically classify incoming customer emails by intent and generate appropriate responses, eliminating manual triage and reducing routing errors.

Competitions/Hackathons

IMC Prosperity 4 | *Python, Pandas, Seaborn, Matplotlib*

Apr 2026

- Engineered and optimized algorithmic trading strategies that increased simulated portfolio returns by 17-29% across multiple trading rounds through systematic signal generation and portfolio rebalancing.

Technical Skills

Stacks: Python, R, C, C++, Java, HTML/CSS, SQL

Machine Learning & AI: Scikit-learn, PyTorch, TensorFlow, Keras, NLTK, SpaCy, LightGBM, XGBoost, OpenCV

Data Science & Analytics: Pandas, Numpy, Seaborn, Scipy, Matplotlib, Pyspark, Statsmodels, Excel, PowerBI, Tableau

Databases: PostgreSQL, MySQL, Microsoft SQL Server, SQLite, Oracle Database, SQLAlchemy

Tools & Platforms: Git, GitHub, Anaconda, Jupyter Notebook, Streamlit, Hugging face, Docker, AWS, Azure, GCP